

Generated Question Paper

Q1. (c) Complete Table 2.1 by stating two structural features of epidermis and mesophyll tissues and explaining how each feature is an adaptation for photosynthesis. [4 marks]

Q2. (iii) State the name of the gas that forms to create the gas bubbles in the dough. [1 marks]

Q3. (d) Scientists had higher breathing rates in the 550 ppm CO₂ plot. [2 marks]

Q4. A: Blood vessel C. Blood vessel C transports blood from the to the . [2 marks]

Q5. (b) State the organs that make up the central nervous system. [1 marks]

Q6. (b) Soybean plants, *Glycine max*, were grown in two plots with different CO₂ concentrations, 370 ppm and 550 ppm. Fig. 2.1 shows average photosynthesis rates. Describe and explain the effect of carbon dioxide concentration on the average rates of photosynthesis of the soybean plants from 04:00 to 22:00. [6 marks]

■ Diagram not found: images/photosynthesis_rate_graph.png

Q7. (d) shows a photomicrograph of a sperm cell. begincenter endcenter noindent (e) Draw and label one human egg cell, including at least one feature not found in a sperm cell. [3 marks]

Diagram not found: images/sperm_cell_micrograph.png

Q8. (e) shows a synapse and parts of two neurons. begincenter endcenter (i) Draw an arrow to show the direction of the signal across the synapse. [1 marks]

[■ Diagram not found: images/synapse_diagram.png]

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..... [1 marks]

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Q10. (ii) Enzymes are proteins. State the names of the chemical elements found in all proteins. [2 marks]

Q11. (a) Describe the possible causes of increased atmospheric carbon dioxide. [3 marks]

Q12. (ii) State the names of the structures labelled A and B on Fig.2. [2 marks]

Q13. Enzymes are catalysts. (a) (i) Describe what is meant by the term catalyst. [2 marks]

Q14. (ii) State one recommendation, other than regular brushing, for proper care of teeth. [1 marks]

Q15. (b) (i) State the name of the source of energy used by organism A. [1 marks]

Q16. (e) Explain the effects of nitrate ion deficiency on plant growth. [3 marks]

Q17. (iii) Ancestors of monkey flower plants had a much smaller leaf area. Explain how the monkey flower plants have developed a larger leaf area over time. [5 marks]